

# EUGENE XUE JIA WEI

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## EDUCATION

University of Adelaide, Australia  
Bachelor of Computer Science (Hons)

Feb 2021 – Feb 2025

- GPA: 6.71/7.00
- Recipient of the Global Citizens Academic Scholarship (15% Scholarship for Undergraduate Studies)
- Relevant Coursework: Data Structures and Algorithms, Artificial Intelligence, Operating Systems, Distributed Systems, Applied Machine Learning

## PROFESSIONAL EXPERIENCE

Jireh Group  
AI Engineer

Apr 2025 – Current  
Singapore

- Improved model **performance from 78% to 87%** by training and evaluating state-of-the-art machine learning architectures to identify the most effective solution for medical image classification use case
- Developed a multi-modal inference pipeline that combines binary classification with multi-class object detection, improving **minority class recall from 24% to 72%** and enable stronger detection of lower-represented cases
- Reduced performance bias across minority classes by calibrating prediction thresholds and evaluating per-class precision and recall, **narrowing recall gap between majority and minority class from 48% to 15%**
- Introduced synthetic datasets using generative AI to balance the minority data scarcity, increasing underrepresented class samples by 15% and boost detection robustness
- Deployed trained model into a production-ready API, enabling batch evaluation up to 100k medical images and scalable processing, reducing **evaluation workflow from 8+ hours of manual review to under 4 hours through automatic batch inference to complement medical professional review.**

Things Smart Pte Ltd (Joint Project with Advanced Micro Foundry)  
Software Developer Intern

Dec 2023 – Feb 2024  
Singapore

- Increase **optimization runtimes by 2x** by adapting genetic algorithm from Python to C, enabling efficient parameter for optical hardware
- Collaborated with Advanced Micro Foundry (AMF) to analyze large voltage-response datasets to determine an optimal analog voltage range for photodetection, **reducing manual parameter tuning time by approximately 40% through batch processing using the algorithm**
- Validated outputs through testing across multiple illumination levels and voltage sweeps, improving detection signal stability, minimizing **measurement variance by 25% through automation**
- Designed a user-friendly dashboard with React and Material UI to visualize complex data outputs in real-time and improve useability for technical stakeholders

## PROJECTS & HACKATHONS

Frontend Developer, AskCodey – AI-Powered Coding Education Platform (NUS GURU AI Innovation Challenge)

Jan 2025 – Apr 2025

- Developed a full-stack technical interview platform, using Next.js, TypeScript, Vercel, featuring an integrated code editor, Leetcode-styled question viewer and LLM-powered coaching assistant
- Engineered scalable UI/UX flows and state management patterns to ensure UI consistency across user workflows and reduced cognitive load
- Boost perceived responsiveness and user engagement with adding micro-interactions and transitions to enhance the visual experience, and optimized PostgreSQL data fetching
- Video Presentation:** [https://www.youtube.com/watch?v=AQkmkYWz\\_CM](https://www.youtube.com/watch?v=AQkmkYWz_CM)

Honours Research, Multi-class Classification of Hip OA Severity

Feb 2024 – Oct 2024

- Implemented a deep-learning framework to achieve multi-class classification of hip osteoarthritis severity, achieving an 82% diagnostic accuracy and 95.6% one-neighbor accuracy
- Address data scarcity with synthetic data generation and ensure robust model performance across diverse datasets
- Designed an intuitive web interface for medical professionals to view real-time severity predictions, along with confidence scores
- Documented research findings and methodology in a comprehensive dissertation as part of the Honours final year project, identifying future pathways for integrating multimodal models into clinical systems

Social Event Scheduler

Mar 2022 - Jun 2022

- Developed a full-stack social web application for students, supporting various social media authentication, club registration, RSVP for events, and manage time zone preferences
- Implemented with Node.js, ExpressAPI on the backend and AJAX on the frontend
- Designed and integrated a database schema to support core application features and enable reliable user and event management workflows
- Applied Agile practices, contributing to sprint planning, retrospectives in a team of 3, to ensure iterative feature delivery and refinement of code quality through modular testing of components within the app framework over 10 weeks

## SKILLS

Python, C++, TypeScript, React, Next.js, MaterialUI, AJAX, Node.js, PostgreSQL, schema design, Vercel, Git, Computer Vision, Generative LLM (LangChain)